

BATESVILLE, AR, USAWMO: **723448**Lat: **35.726N**Lon: **91.647W**Elev: **141**StdP: **99.64**Time Zone: **-6.00 (NAC)**Period: **94-19**WBAN: **53934****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.9	-6.8	-17.3	0.8	-5.7	-14.2	1.1	-3.3	8.7	11.1	7.8	10.7	1.9	320	0.406

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	36.3	24.6	34.8	24.5	33.2	24.2	26.4	33.0	25.7	32.3	25.2	31.7	3.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.3	19.6	29.9	23.9	19.0	29.4	23.0	18.0	28.4	82.8	33.2	80.3	32.4	77.7	31.7	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
7.5	6.5	5.6	DB	-12.7	38.2	2.3	2.1	-14.4	39.7	-15.7	41.0	-17.0	42.2	-18.6	43.7	
			WB	-13.5	27.4	2.1	1.0	-15.0	28.1	-16.2	28.6	-17.4	29.2	-18.9	29.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.1	4.2	6.3	10.8	16.1	20.7	25.2	27.1	26.7	22.6	16.6	10.3	5.6
	DBStd	9.34	6.08	5.86	5.70	4.59	3.98	2.93	2.71	2.89	3.78	4.66	5.09	5.41
	HDD10.0	617	198	132	59	7	1	0	0	0	0	5	57	158
	HDD18.3	1877	439	338	241	96	24	1	0	0	9	90	244	396
	CDD10.0	2827	17	28	84	188	332	455	530	518	378	209	67	21
	CDD18.3	1047	1	1	7	27	97	206	272	261	137	36	3	1
	CDH23.3	10722	1	4	45	210	801	2140	3048	2831	1311	311	21	1
	CDH26.7	4759	0	0	3	34	228	948	1501	1418	547	78	2	0
Wind	WSAvg	2.4	2.7	2.8	3.0	3.0	2.4	2.1	1.9	1.8	1.8	2.1	2.4	2.5
Precipitation	PrecAvg	1269	105	96	124	117	123	98	97	81	82	95	118	121
	PrecMax	1769	213	184	268	286	258	286	189	170	220	305	313	257
	PrecMin	681	38	36	34	7	18	30	25	19	0	4	9	8
	PrecStd	262	41	42	57	60	61	54	44	43	53	66	84	65
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.3	23.7	27.2	29.1	32.4	36.4	37.9	38.7	36.2	31.3	26.2	21.3
		MCWB	16.2	16.7	17.8	20.1	22.7	24.4	24.6	24.3	23.0	21.4	18.2	17.2
	2%	DB	18.7	20.9	24.1	27.6	30.9	34.1	36.2	37.0	33.7	28.8	22.7	18.7
		MCWB	15.1	15.5	16.6	18.6	22.5	24.0	25.1	24.5	23.1	20.4	16.5	15.8
	5%	DB	16.4	17.9	22.3	26.1	29.0	32.8	34.1	34.8	32.2	27.1	21.1	17.0
		MCWB	12.1	13.3	15.5	17.9	21.8	23.8	24.9	24.4	22.5	18.9	16.1	13.5
	10%	DB	13.8	16.1	20.0	23.9	27.6	32.1	33.0	32.9	30.0	24.5	18.7	13.9
		MCWB	9.9	11.1	14.1	17.0	20.8	23.6	24.9	24.0	21.9	18.0	14.6	10.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.8	18.5	19.6	21.7	24.7	26.6	27.4	27.3	25.5	23.5	20.2	18.2
		MCDB	19.8	22.0	25.0	27.2	30.0	33.2	33.6	33.7	32.1	28.4	23.7	20.2
	2%	WB	16.4	16.8	18.1	20.5	23.6	25.5	26.6	26.3	24.6	21.9	18.5	16.5
		MCDB	18.3	19.2	22.8	25.5	28.7	31.8	33.2	32.9	30.9	26.2	21.4	18.3
	5%	WB	13.6	14.6	16.7	19.5	22.7	24.8	25.9	25.6	23.8	20.5	16.8	13.9
		MCDB	15.7	17.7	21.0	24.1	27.7	31.0	32.5	32.3	29.6	24.6	19.9	16.3
	10%	WB	10.3	11.8	15.2	18.4	21.8	24.2	25.4	24.9	22.9	19.2	14.9	11.4
		MCDB	13.4	14.9	19.1	22.6	26.5	30.1	32.0	31.4	28.1	23.5	18.3	13.6
Mean Daily Temperature Range	5% DB	MDBR	9.7	10.1	10.9	11.6	10.9	11.1	11.0	11.6	12.3	12.1	10.6	9.2
		MCDBR	12.4	13.0	12.8	13.1	11.9	12.4	12.1	13.5	14.2	14.2	12.6	11.4
		MCWBR	9.3	9.1	7.6	7.0	5.8	5.0	4.5	4.5	5.6	7.3	8.2	8.5
	5% WB	MCDBR	10.4	11.2	11.1	10.2	10.1	10.5	10.9	11.3	12.0	11.5	10.6	10.2
		MCWBR	9.5	9.3	7.8	6.6	5.5	4.9	4.6	4.7	5.3	6.7	8.4	8.8
Clear-Sky Solar Irradiance	taub	0.315	0.322	0.357	0.391	0.431	0.455	0.486	0.474	0.410	0.355	0.324	0.308	
	taud	2.485	2.463	2.398	2.299	2.249	2.197	2.159	2.168	2.339	2.473	2.511	2.520	
	Ebn at Noon	886	920	912	893	858	833	805	806	846	871	868	872	
	Edh at Noon	83	95	111	129	138	145	150	146	117	94	81	76	
All-Sky Solar Radiation	RadAvg	2.43	2.98	4.10	5.18	5.70	6.52	6.23	5.73	4.88	3.73	2.64	2.09	
	RadStd	0.21	0.39	0.31	0.37	0.45	0.44	0.31	0.37	0.37	0.46	0.28	0.23	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days	
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